David Andrei-Gabirel

+40760230872 | d.andreeei20@gmail.com

linkedin.com/in/andrei-david-2b7538357/ | github.com/andreidavid2003 |

Slobozia, Ialomita, Romania

Education

- **B.Sc. in Computer Science**

  Oct 2022 – 2026 (expected)

  - Faculty of Automatic Control and Computer Science, University of POLITEHNICA of Bucharest.
  - Relevant coursework: Object-Oriented Programming, Data Structures and Algorithms, Operating Systems, Artificial Intelligence, Communication Protocols, Parallel and Distributed Algorithms, Database.

- **Graduated Mathematics-Information Specialization**

  Sept 2018 – May 2022

  - "Mihai Viteazul" National Collage Slobozia

Projects

- **Command-Based Tape Editor Simulator**

  C []

  - Developed a C-based tape editor using custom data structures and pointer logic.
  - Implemented command execution with undo/redo via stacks and a queue system.
  - Ensured memory safety and error handling through manual allocation and cleanup.

- **eBanking App**

  Java []

  - Developed an eBanking App featuring user connectivity and personalized stock recommendations.
  - Enhanced data processing by integrating CSV file reading.
  - Improved reusability, scalability and maintainability by applying design patterns

- **HTTP, Client Rest**

  C []

  - Used the HTTP protocol and JSON concepts.
  - Worked with RESTful APIs, and JWT for secure authentication.
  - Interacted with JWT and cookies for user session management and tracking user activity.

- **Regex-NFA Conversion**

  Python []

  - Implemented Thompson's construction to convert regex patterns into NFAs with epsilon transitions.
  - Supported standard regex operators (|, ., *, +, ?) and predefined ranges ([a-z], [A-Z], [0-9]).
  - Built a recursive descent parser for regular expressions to produce composable NFA structures.

- **NFA-DFA Conversion**

  Python []

  - Implemented NFA to DFA conversion via subset construction with epsilon-closure handling.
  - Designed a DFA minimization algorithm using partition refinement (Hopcroft's method).
  - Modeled automata transitions and states using generic dataclasses and Python type hints.

Technical skills

- **Programming Languages:** C, C++, Java, Linux, SQL (intermediate), Python (beginner), Shell Scripting, Assembly ARM x86
- **Developer Tools:** VS Code, IntelliJ IDEA, Visual Studio